

DOUBLY LINKED LIST

PROGRAM CODING:

```
#include<conio.h>

#include<stdio.h>

struct doubly
{
    struct doubly *front;
    struct doubly *back;
    int i;
};

void main()
{
    struct doubly *head=0;
    struct doubly *last=0;
    struct doubly *p=0;
    struct doubly *temp=0;
    int ch,intch,user,cnt=0,t,add;

    clrscr();

    printf("\n\t 1. CREATE.");
    printf("\n\t 2. INSERT.");
    printf("\n\t 3. DELETE.");
    printf("\n\t 4. DISPLAY.");
    printf("\n\t 5.EXIT.");
    scanf("%d",&ch);
```

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while(ch!=5)
{
    if(ch==1)
    {
        cnt=0;

        head=(struct doubly *)malloc(sizeof(struct doubly));

        head->back=0;

        head->front=0;

        printf("\n\t ENTER DATA::: ");

        scanf("%d",&head->i);

        cnt++;

        last=head;
    }

    if(ch==2)
    {

        printf("\n\t 1. INSERTION AT FRONT.");
        printf("\n\t 2. INSERTION AT MIDDLE.");
        printf("\n\t 3. INSERTION AT END.");

        scanf("%d",&intch);

        temp=(struct doubly *)malloc(sizeof(struct doubly));

        printf("\n\t ENTER DATA:::");

        scanf("%d",&temp->i);

        if(intch==1)
        {

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temp->front=0;

temp->back=head;

head=temp;

cnt++;

}

if(intch==3)

{

    p=head;

    while(p->back!=0)

    {

        p=p->back;

    }

    temp->back=0;

    temp->front=p;

    p->back=temp;

    cnt++;

}

if(intch==2)

{

    printf("\n\tENTER VALUE BETWEEN 1--%d",cnt);

    scanf("%d",&add);

    t=1;

    p=head;

    while(t<add)

    {

        p=p->back;

        t++;

    }

}

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    }

    temp->back=p->back;

    p->back->front=temp;

    p->back=temp;

    temp->front=p;

}

}

if(ch==4)
{
    p=head;
    while(p!=0)
    {
        printf("\t-->%d",p->i);

        p=p->back;
    }

    printf("\n\t total nodes %d",cnt);
}

if(ch==3)
{
    printf("\n\t1. DELETE FRONT");

    printf("\n\t2. DELETE MIDDLE");

    printf("\n\t3. DELETE END");

    scanf("%d",&intch);

    if(intch==1)
    {

```

```

        head->back=t;

        free(head);

        head=t;

        head->front=0;

        cnt--;
    }

    if(intch==3)
    {
        p=head;

        while(p->back!=0)
        {
            p=p->back;
        }

        temp->front=p;

        p->back=temp;

        temp->back=0;

        cnt--;
    }

    if(intch==2)
    {

        printf("\n\tENTER VALUE BETWEEN 1--%d",cnt);

        scanf("%d",&add);

        t=1;

        p=head;

        while(t<(add-1))
        {

            p=p->back;

```

```
        t++;  
    }  
    temp=p->back;  
    p->back=temp->back;  
    temp->back->front=p;  
    free(temp);  
  
    }  
  
    }  
    printf("\n\t 2. INSERT.");  
    printf("\n\t 3. DELETE.");  
    printf("\n\t 4. DISPLAY.");  
    printf("\n\t 5.EXIT.");  
    scanf("%d",&ch);  
}  
getch();
```